運用水下滑翔機於深海海域觀測內孤立波

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摘　　要

南海是內孤立波(Internal Solitary Wave, ISW)盛行的海域，其源自潮流越過呂宋海峽內的蘭嶼海脊與恆春海脊而產生內波並往西的傳遞，之後因非線性效應逐漸變為內孤立波、且振幅也逐漸增大，最大振幅可達 200 公尺，是全世界最大的觀測紀錄。當內孤立波傳遞至大陸斜坡與大陸棚海域時，因地形淺化而波型變陡、翻轉、破碎，最後消散(Alford et al., 2011, 2015; Chang et al., 2021; Echeverri & Peacock, 2010; Farmer et al., 2011; Huang et al., 2016; Lien et al., 2014; Ramp et al., 2019)。巨大振幅的內孤立波會產生強烈的水平流場垂直流切、迅速的上下起伏與水文變化，進而會影響水下通訊與潛水艇航行安全(Apel et al., 1997; Colosi and Lynch, 1999; Katz, 1967; Lynch, et al., 1996; Rubenstein, 1999)，是潛艇作戰/反潛作戰的重要議題之一。

觀測內孤立波的方法計有運用海流儀、溫度計串、衛星、船載雷達等方法，但都有其限制。例如放置於水下錨定的海流儀或是溫度計串，無法近即時回傳觀測資料、衛星觀測受限於天氣、船載雷達則無法達到大範圍觀測。水下滑翔機(underwater glider)是一種利用阿基米德原理運作的無動力自主式觀測平台。它透過調整自身的體積、浮力、重心位置與姿態，在無需推進器的情況下即可實現自主下潛、上浮與滑翔至指定位置。在滑翔過程中，水下滑翔機能持續收集沿途的海洋水文資料，並於浮出水面後，將資料上傳至地面控制中心。由於具備低成本、高效率與廣泛應用等優勢，是執行海洋水文觀測任務的潛力工具。本研究乃是擬運用水下滑翔機於深海域所觀測的近即時資料，估算內孤立波振幅與波速。

國立臺灣大學研究團隊曾經發展運用水下滑翔機近即時觀測資料估算內孤立波振幅與波速(Yang et al., 2023)，其方法是運用內孤立波通過時對水下滑翔機所造成的迅速垂直位移(如圖 1 所示)，估算內孤立波的垂直速度，然後再運用水下滑翔機所觀測到的近即時水文資料並結合 Dubreil–Jacotin–Long (DJL, Stastna & Lamb, 2002)模式，估算內孤立波振幅與波速。此方法的運用只限於水深淺於水下滑翔機可達到深度的海域，若是海域深度超過水下滑翔機可達到的深度，則無水下滑翔機可達深度以深的水文資料，進而無法解算 DJL 模式。為了解決此一限制，我們蒐集了深海域的歷史水文資料，統計深海域的四季平均之密度垂直剖面分布，並結合水下滑翔機所測得的近即時上層海洋水文資料，並將其運用至 DJL 模式計算內孤立波振幅與波速。以圖 1 的觀測內孤立波案例，計算所得的振幅與波速分別為 126 m 與 299 cm/s。比對運用向日葵 8 號衛星影像估計所得之波速為 303 cm/s，兩者波速非常接近，證明此方法之可行性。詳細的研究成果將展示於研討會之海報上。

關鍵詞：內孤立波、水下滑翔機、反潛作戰

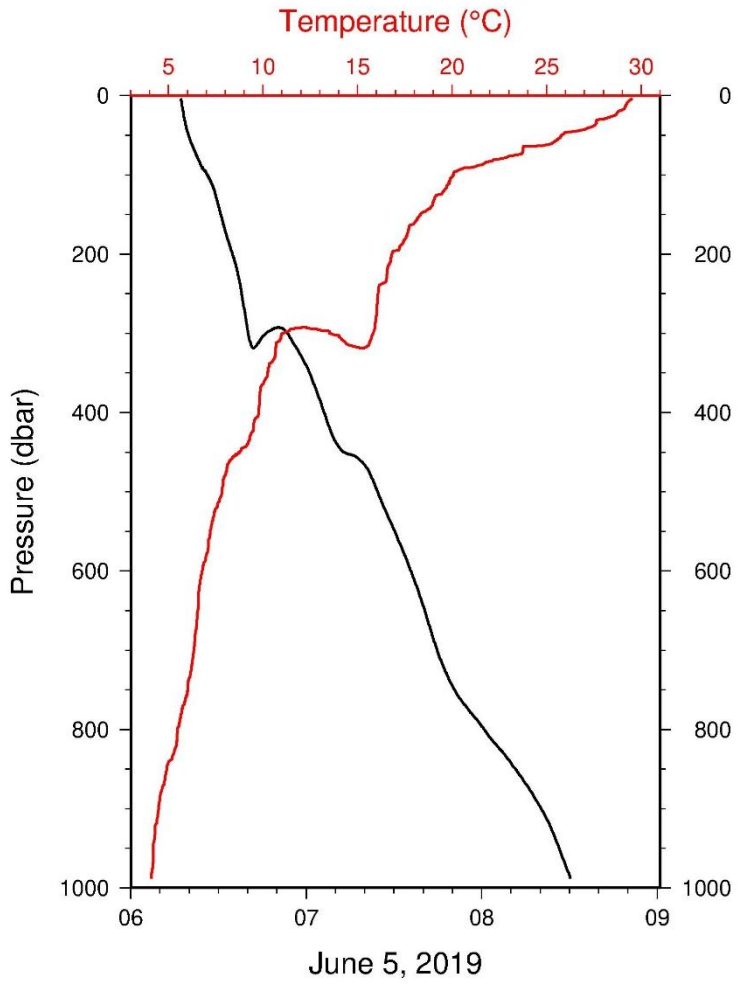


圖 1: 水下滑翔機所觀測到內孤立波之案例。

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